

Early Detection of AD/ADRD: A Robust, Fair, and Explainable Framework for Cognitive Score Prediction

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1. ABSTRACT

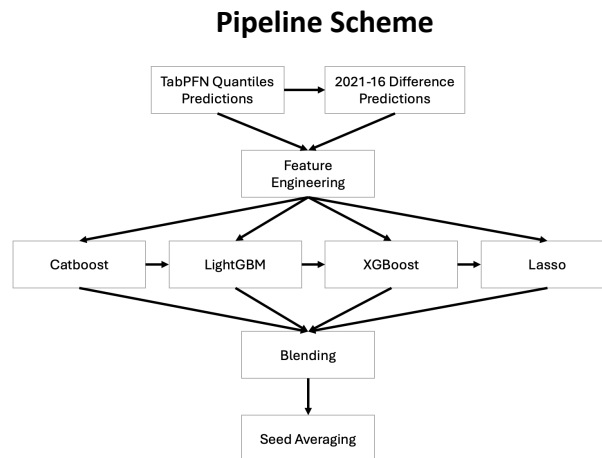
In recent years, machine learning (ML) has made significant advances in large-scale data tasks. However, applying these successes to smaller-scale datasets remains challenging. Small tabular datasets, common in many real-world applications, present unique difficulties, including complex feature interactions, data heterogeneity, and subgroup-specific patterns. To address these challenges, we developed a pipeline for this competition that integrates diverse models with domain-specific insights.

A key innovation in our approach was leveraging **TabPFN**, a pre-trained classification model adapted to predict quantiles of the target variable, which enriched the feature space and boosted the performance of other models in the pipeline. Additionally, the predicted **2016-2021 relative difference in composite scores** was incorporated as a feature in the downstream pipeline, which improved our models' ability to capture temporal trends. Notably, we implemented a **sequential prediction framework** using gradient-boosted ensemble models, where each model iteratively refined predictions based on prior outputs. To ensure robustness and stability, we finally employed **age-group blending** and **multiple-seed averaging**. This comprehensive strategy secured us 1st place on both the public and private test leaderboards, with overall test RMSE=38.19.

Our findings highlighted several factors significantly associated with changes in cognition up to 18 years in advance. Improved cognition was predicted by factors such as deeming religion unimportant, unmet life goals, younger age, and higher education, while worsened cognition was linked to older age, poor memory, and depression. Furthermore, engaging in activities like puzzles, reading, and volunteering predicted cognitive scores up to 1.5% higher than non-engagement over periods of 4 and 9 years. Cognition was measured using a composite score specified by the competition rules, where lower scores indicated increased risks of Alzheimer's disease or Alzheimer's-related dementias (AD/ADRD).

2. METHODOLOGY

Pipeline Overview and Justification of Modeling Choices



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Our pipeline incorporated diverse models and techniques to tackle the unique challenges of small tabular datasets.

1. TabPFN is a Transformer-based model optimized for small tabular datasets. It uses Prior-Data Fitted Networks (PFNs) pre-trained on synthetic datasets generated from Bayesian priors and causal models, such as Directed Acyclic Graphs, simulating realistic data processes (e.g., linear/non-linear interactions, feature dependencies, and class imbalances). This pre-training enables TabPFN to generalize to unseen datasets without fine-tuning [1]. By incorporating Bayesian principles and causal reasoning, it provides robust, interpretable predictions while avoiding spurious correlations [2]. TabPFN delivers single-pass predictions without iterative optimization but supports only up to 100 features. For this project, we used CatBoost feature importance rankings to select the top 100 features, ensuring relevance and efficiency.

For our task, TabPFN was adapted to predict 10 quantiles of the cognitive ability score. These quantiles provided probabilistic insights into the target variable, enhancing the feature space for downstream models.

2. Feature Engineering

- **Dummy Variables:** Applied one-hot encoding to all categorical variables to create dummy variables for model training.
- **Ordinal Variables:** Used ordinal encoding for variables where the order carried significance, such as global health, BMI, and life satisfaction.
- **Log Transformation:** Applied natural logarithms to all income variables to normalize their distributions and reduce skewness.
- **Pseudo-Numeric Variables:** Converted variables with ranges, such as education levels and the number of living children, into pseudo-numeric values by using their midpoints. This simplified processing while retaining meaningful variability.
- **Higher-Order Terms:** Created squared terms for age and income, as well as interaction terms between these variables, to capture non-linear and combined effects.
- **Change Over Time:** Added absolute differences between all numeric variables, including dummy-coded ones, across 2003 and 2012 to capture temporal dynamics. This aligns with research linking negative wealth shocks [3] or a history of depression [4] to cognitive decline and rural-to-urban transitions to cognitive benefits in later life [5].
- **Survey Participation Type:** Introduced a variable to represent the pattern of survey participation across different years, with six distinct categories based on participation patterns (e.g., [2003, 2012, 2021] → Category 1, [2012, 2016] → Category 6).
- **Advanced Statistics by Groups:** Computed group-level statistics (e.g., min, max, mean, median, SD, skewness, range, IQR, MAD, CV, Gini, entropy, harmonic mean, geometric mean, peaks, troughs) for the composite score using the training set. Groups were defined based on key features such as 2012 education, 2012 age, and survey participation type. These statistics were transferred to the test set from the train set to prevent data leakage.

3. 2016-2021 Relative Difference Prediction: A LightGBM model was used to predict the relative difference in composite scores between 2016 and 2021, leveraging all the above features. The model was trained on participants with data from both years and applied to predict the relative difference for all observations. These predictions were then incorporated as features in the subsequent models predicting the original composite score.

4. Base Models: Multiple models were trained, including Gradient Boosted Trees (Xgboost, CatBoost, and LightGBM) and Lasso Regression. These models provided a mix of non-linear and linear perspectives, ensuring diversity and complementarity.

5. Sequential Prediction: Predictions on the test data were generated sequentially, starting with LightGBM and using the outputs of each model as features for subsequent models. This sequence concluded with Lasso Regression, which leveraged refined predictions to capture linear relationships more explicitly. Sequential modeling has been shown to be effective in capturing cumulative improvements in predictive accuracy, particularly for complex datasets with interdependent feature relationships.

6. Group-Based Blending: Predictions from all models were blended using domain-specific groupings, such as age ranges (age was selected since it is the strongest predictor of cognition) and quantiles of the composite score predictions generated by the four mentioned models. Group-based blending has been demonstrated to improve predictive performance by refining outputs for specific subgroups, particularly when domain knowledge is integrated into the grouping strategy (like age in our case). The top 20 largest groups were identified using the training set, and predictions within each group were weighted based on their RMSE performance with a SoftMax function.

7. Multiple-Seed Averaging: To enhance robustness, the entire pipeline was repeated across multiple random seeds, generating a distribution of predictions. The final output was the average of predictions from all seeds. This approach ensures stability and reduces sensitivity to random initialization.

Model Training, Hyperparameter Tuning, and Validation

Avoiding Data Leakage: Out-of-fold (OOF) predictions instead of training data predictions were used to generate features. We ensured that each training instance had a corresponding prediction from a model trained without that instance. OOF predictions were used for features such as TabPFN-generated quantiles, LGBM-predicted relative differences in composite scores, and main model predictions for the sequential predictions stage.

Cross-Validation Strategy: We implemented a group-based cross-validation (CV) strategy to ensure reliable evaluation and prevent data leakage. Specifically, we combined 5-fold splits from the TabPFN training with 5-fold splits from the relative difference prediction model, resulting in a total of 25 unique folds. This approach was essential for avoiding data leakage, as the 2016-2021 composite score difference prediction used only a subset of the dataset for training – cases where both 2016 and 2021 measures were available.

Hyperparameter Optimization: We employed Optuna for hyperparameter tuning [6] with 5 trials to balance computational efficiency and minimize overfitting. For the Lasso model, missing values were imputed using the best-performing quantile optimized via Optuna.

Generalizability

The generalizability of our approach can be demonstrated by the consistent performance of the final model across all cross-validation (CV) folds. For brevity, we summarized the RMSE performance of our final model for the two CV schemes mentioned above – a 5-fold CV for predicting TabPFN features and another 5-fold CV for predicting the 2016-2021 composite score difference, both used as features in our downstream models. For predicting TabPFN features, the Mean RMSE was 42.47 (SD=1.58), while for predicting the 2016-2021 composite score difference, the Mean RMSE was 30.92 (SD=3.20). The relatively low SDs in both cases indicate consistent fold-level performance, underscoring the generalizability of our model. The overall CV RMSE on the train set was 37.97, relatively closely matching the RMSE observed on the test set (38.19).

A moderate limitation to the generalizability of our methodology is the dataset's unique longitudinal structure. Other studies may have varying time intervals, more or fewer waves, or different assessment frequencies, directly impacting our change-over-time features and result interpretation.

Robustness

To ensure robustness, training and prediction processes were repeated using multiple random seeds and predictions were averaged across the seeds. The RMSE showed slight variability across the seeds (Mean=38.27, SD=0.05), highlighting the importance of accounting for this variation to enhance prediction reliability. In addition, the SD of seed-specific predictions per individual was computed, with an average SD of 3.41. The presence of this variability suggests that averaging predictions across seeds can improve robustness compared to relying on a single random seed.

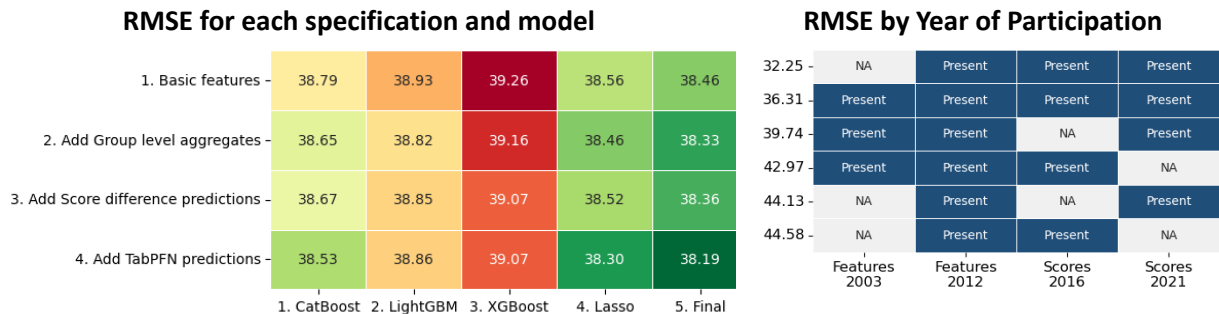
To further evaluate robustness, perturbation analyses were conducted by systematically modifying the data. Two scenarios were tested: 1. Removing outliers in all income variables (defined as values above the 99th percentile) 2. Eliminating observations with more than 50% of missing values in 2003 and 2012. In both scenarios, RMSE remained stable (38.27 and 38.26, respectively), demonstrating our approach's resistance to data preprocessing adjustments and confirming its reliability under changed conditions. Finally, we randomly removed 5% of the features and RMSE=38.59 remained close to that of the full test data (38.19), further confirming the model's resilience to distribution shifts.

3. PERFORMANCE ANALYSIS

The heatmap on the left-hand side below shows the performance of our consecutive model specifications. The RMSE metric was chosen because it is required for the competition and seems crucial for predicting cognitive scores, as it emphasizes large errors, enabling accurate detection of significant cognitive changes essential for timely interventions.

The key takeaway is that increasing model complexity – by incorporating advanced group-level aggregates, composite score difference predictions, and TabPFN quantiles as features – consistently improves RMSE performance in most cases. Additionally, performance improved across model types at the stage where sequential predictions were generated, leading to further enhancements in the final model's RMSE. Notably, the final model (progressing through CatBoost → LightGBM → Xgboost → Lasso → Final) achieved better performance, even though intermediate models, such as Xgboost, could perform slightly worse than the models they rely on (e.g., CatBoost and LightGBM).

The accompanying table on the right-hand side shows RMSE for the 6 groups of people with specific combinations of survey participation years. The results reinforce our earlier observation from the fold-specific analysis: models behaved better with data from multiple participation years.



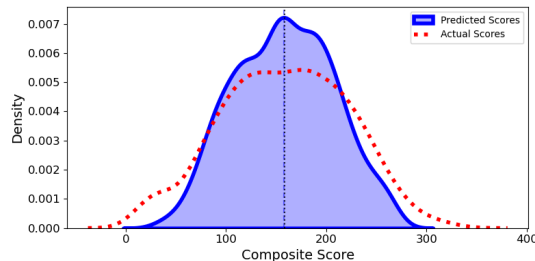
Further, an error analysis was carried out by calculating RMSE for the final model across different quantiles of the composite score and visualizing predicted vs. actual values. The results show better performance in the middle of the distribution, with a tendency to underpredict higher values and overpredict lower values across all model specifications. This tendency might occur due to the model's optimization process focusing on minimizing overall error, which can lead to underfitting at the extremes of the distribution where data is often sparse or exhibits higher variability. Nevertheless, our most

advanced model demonstrates reduced RMSE at the tails of the distribution, indicating improved performance in those regions.

RMSE for each Specification and Scores Quantile

1. Basic features	49.33	32.47	31.64	30.95	43.71
2. + Group level aggregates	49.06	32.81	31.53	30.63	43.48
3. + Score difference predictions	48.97	32.78	31.70	30.84	43.47
4. + TabPFN predictions	48.79	33.05	31.86	30.25	43.05
	Q1	Q2	Q3	Q4	Q5

Density of Predicted vs Actual Scores



4. BIAS AND ETHICS

Dataset Bias

The dataset for this challenge had quite a unique longitudinal format with four waves: the first two (2003, 2012) capture features, and the latter two (2016, 2021) measure outcomes. There were six distinct combinations of survey participation, reflecting varied respondent patterns (see table below). Therefore, to evaluate potential biases, we analyzed the entire dataset (train and test data combined) for systematic differences in socio-economic features across these participation types.

This table highlights disparities in representation based on participation years. For example, rural residents – linked to health disparities in dementia [7] – were overrepresented among those who participated in [2012, 2016, and 2021], or [2012 and 2021], indicating selection bias. Additionally, the 80+ age group was underrepresented among the cohort of people who have 2 measures of cognitive abilities in 2016 and 2021, pointing to attrition bias likely caused by mortality [8]. To address this, we included all these covariates in our models to mitigate potential biases.

T-test comparisons between sociodemographic means (proportions) within each participation type and those who participated in all four survey years (first row)

	Age 50-59	Age 60-69	Age 70-79	Age 80+	Men	Rural	Education	Migration
[2003, 2012, 2016, 2021]	0.1	0.55	0.27	0.04	0.36	0.4	4.31	0.1
[2003, 2012, 2016]	0.03***	0.33***	0.4***	0.2***	0.43*	0.36	3.64***	0.12
[2003, 2012, 2021]	0.07***	0.57	0.27	0.05	0.4	0.43	4.08	0.09
[2012, 2016, 2021]	0.79***	0.18***	0.02***	0.0***	0.43*	0.47*	5.68***	0.04***
[2012, 2016]	0.66***	0.23***	0.07***	0.04	0.52***	0.38	5.56***	0.07
[2012, 2021]	0.76***	0.2***	0.02***	0.0***	0.54***	0.5***	5.4***	0.1

Note. The column features were derived from the 2012 wave, during which all participants were included.

Model Bias

We introduced a fairness index (FI) to evaluate bias in the predictions:

$$FI = 1 - \frac{\max(RMSE_{subgroup})}{\text{Range of } RMSE}$$

where $RMSE_{subgroup}$ is the RMSE calculated for specific subgroups within a dataset, such as women or age 80+. This metric directly builds on RMSE, ensuring alignment with the primary evaluation objective, and addresses fairness by quantifying subgroup-specific performance disparities, a widely recognized approach [9]. An FI of 1 indicates perfect fairness, where all subgroups have equal RMSE (range=0), while an FI of 0 reflects maximum unfairness, with the minimum RMSE being 0 (range=max RMSE).

Subpopulations tested include categories from age, gender, education, income bins, and urbanity. The results show minimal variability across models, indicating a stable level of fairness

regardless of the model specification. The final model displayed slightly fairer performance compared to some separate models. However, fairness varied across feature groups, with age showing slightly less equitable performance.

Fairness Metrics for each Model and Feature

1. CatBoost	0.92	0.83	0.94	0.88	0.96
2. LightGBM	0.91	0.82	0.96	0.90	0.96
3. XGBoost	0.91	0.81	0.96	0.89	0.96
4. Lasso	0.91	0.82	0.94	0.89	0.95
5. Final	0.91	0.81	0.95	0.89	0.95
6. Age-calibrated fairness	0.91	0.88	0.91	0.89	0.96
	Gender	Age	Education	Income	Urban

Bias Mitigation

To address this, we designed a **customized loss function** for our gradient boosted models that, for example, assigned greater weight to age groups with poorer predictions [10]. This approach improved performance for disadvantaged groups, as shown by the FI for the fairness-adjusted predictions in the heatmap above. However, because this fairness adjustment slightly reduced overall model performance (RMSE=38.95), we offered it as an optional solution for generating fairer predictions but did not include it in our final specification.

Privacy Considerations

In our approach, we developed a method to reconstruct and refine age categories based on a 9-year interval between 2003 and 2012. For individuals with unchanged age categories, their 2003 age was assumed to align with the lower bound of the range. For those with category changes, the 2003 age was estimated as the midpoint between the 2003 lower bound and the 2012 upper bound. New age estimates for 2012 were calculated by adding 9 years to the adjusted 2003 values. The reconstructed ages were integrated into model training and group-based blending.

However, this narrowing of age groups may inadvertently increase the risk of re-identification, as it enhances granularity and makes individuals more identifiable. This issue is particularly critical in medical settings with longitudinal data, where the unique combination of diseases and other attributes could further heighten identifiability risks. Careful attention is required to strike a balance between offering sufficient granularity for accurate predictions and using bin-based variables to obscure sensitive details. Future approaches should weigh these trade-offs to ensure robust modeling while preserving privacy.

5. EXPLAINABILITY AND INSIGHTS

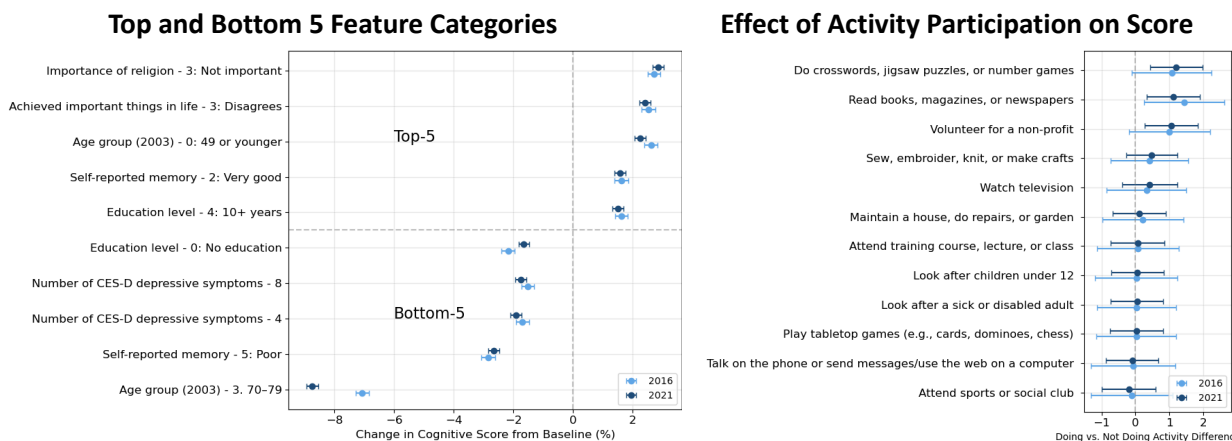
To demonstrate model explainability, two analyses were performed:

Feature importance analysis: We computed the percentage change in composite scores for individuals with specific attributes by fixing a variable to a particular value across the test set (same for both 2003 and 2012 for clarity), predicting scores using the final model, and comparing results with the baseline test set average. Confidence intervals were generated via bootstrapping (see left-hand side of the graph below for the top and bottom 5 features).

This approach, akin to partial dependence analysis [11], revealed that attributes such as viewing religion as unimportant, disagreement with achieving life goals, younger age, and higher education were linked to higher cognitive abilities. Conversely, older age, poor self-reported memory, and depressive

symptoms were associated with lower cognitive abilities, aligning with the existing literature [4], [12], [13].

Counterfactual scenario simulation: This approach also allowed us to simulate potential policy interventions, for example altering the presence of certain activities (e.g., doing puzzles, reading, volunteering) in 2012 test data and calculating predicted cognitive scores. Engaging in these activities improved cognitive abilities 4 and 9 years later by up to 1.5% compared to non-engagement (see right-hand side of the graph below) [8].



6. LIMITATIONS AND FUTURE MITIGATION

Reliance on TabPFN’s Pre-training: TabPFN’s performance depends on the synthetic data used for pre-training, hence potential differences between synthetic and real-world data could impact its effectiveness. Mitigation: Analyze TabPFN’s predictions carefully and compare them with other base models to identify and address potential biases.

Sequential Prediction Sensitivity: The sequential prediction approach refines predictions incrementally but is sensitive to early-stage errors, which subsequent models may propagate. Mitigation: Automated ML pipelines like AutoGluon, which employ diverse models and ensembling strategies, can enhance robustness and reduce error propagation.

Group-Based Blending and Overfitting Risk: Focusing on groups of data during blending may lead to overfitting to specific data patterns. Mitigation: Regularization techniques (e.g., penalizing large weights) or using CV for the group weight selection can reduce overfitting and improve generalization. Age-based grouping remains justified as age is a strong predictor of cognitive abilities. Further domain-specific analysis could uncover additional insights [14].

7. CONCLUSIONS

Generally, there are no severe limitations in terms of using the described approach on other datasets with a longitudinal follow-up structure similar to the one employed in this competition. The potential clinical applications are vast; our analyses could help determine AD/ADRD at early stages, enabling timely interventions and personalized care.

Also, there are great opportunities for further demonstration of our approach on similar datasets with longitudinal structures [15]. To facilitate this, we constructed an automated pipeline with modular classes, enabling seamless testing of the approach on other datasets. We developed a user-friendly routine that can scrutinize multiple aspects of the whole modeling process in terms of errors and biases and extract the effect of SDH through counterfactual simulations. This pipeline greatly enhances the reproducibility and extensibility of our work.

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